

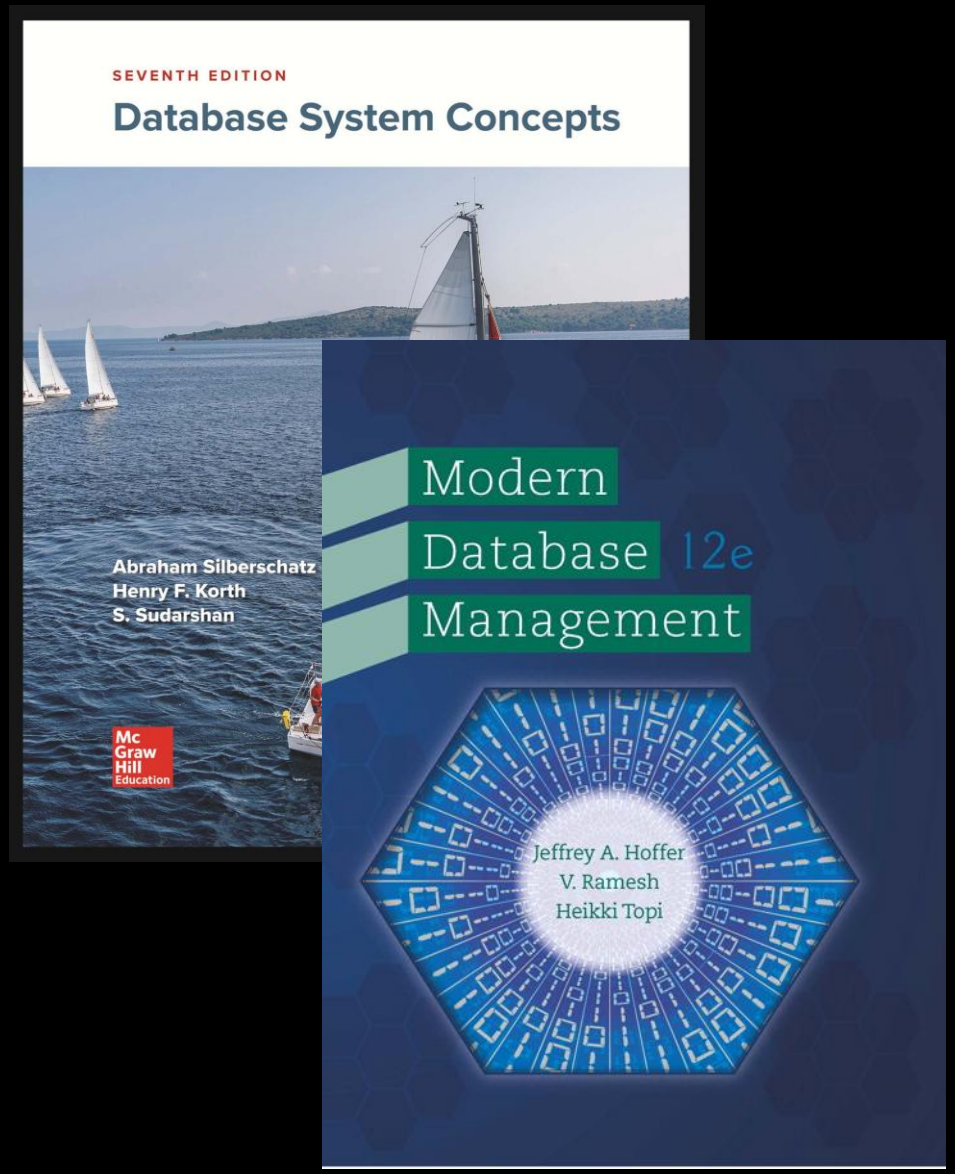
IF2040 – Pemodelan Basis Data

Introduction to Database

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Modified form Silberschatz's slides, Database System Concept, 7th edition

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References

Abraham Silberschatz, Henry F. Korth, S. Sudarshan : "**Database System Concepts**", 7th Edition

- Chapter 1: Introduction

Jeffrey A. Hoffer, Mary B. Prescott, Heikki Topi : "**Modern Database Management**", 12th Edition

- Chapter 1: The Database Environment and Development Process

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Components of the Database Environment

CASE Tools – computer-aided software engineering

Repository – centralized storehouse of metadata

Database Management System (DBMS) – software for managing the database

Database – storehouse of the data

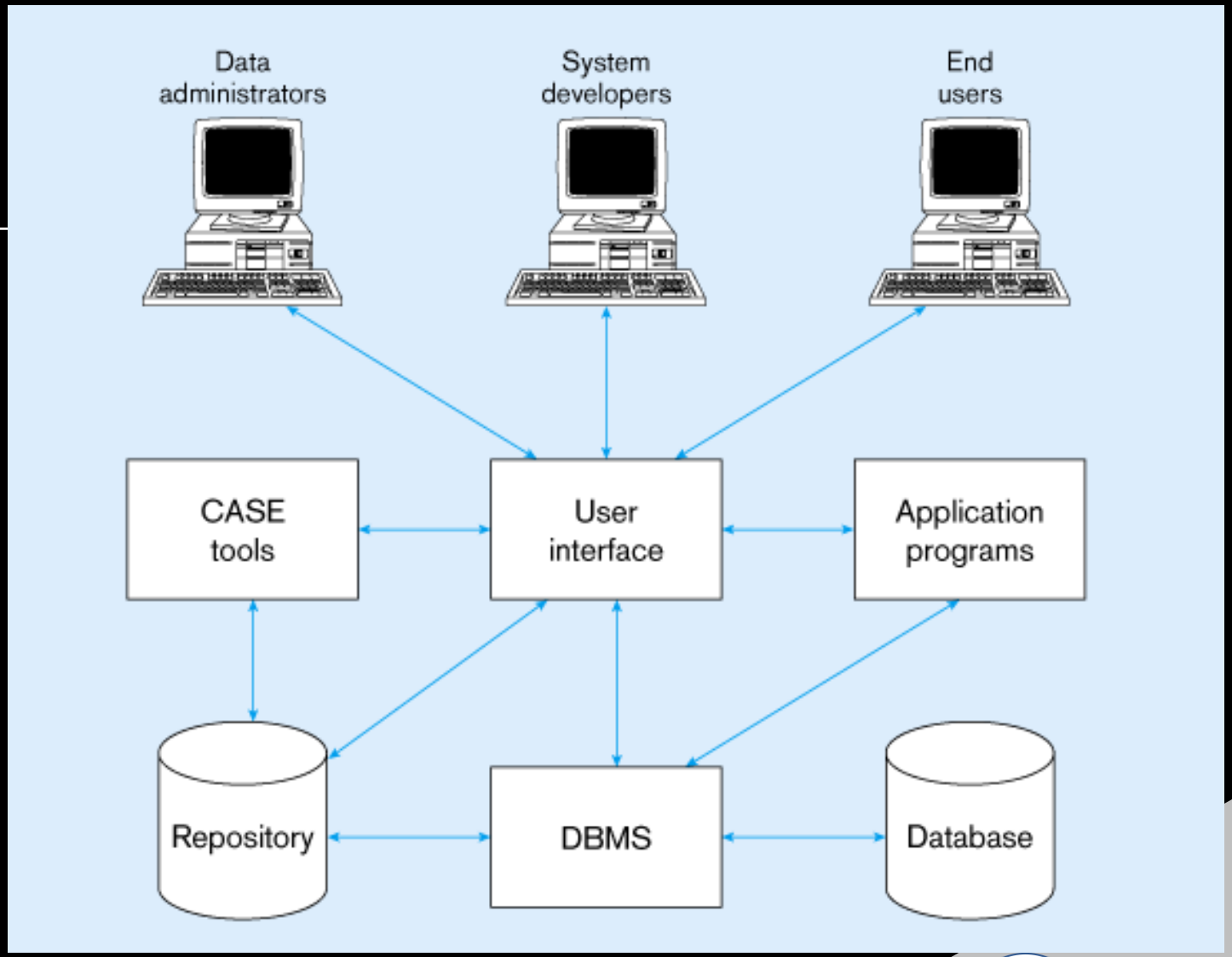
Application Programs – software using the data

User Interface – text and graphical displays to users

Data Administrators – personnel responsible for maintaining the database

System Developers – personnel responsible for designing databases and software

End Users – people who use the applications and databases



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The Range of Database Applications

Personal databases

Workgroup databases

Departmental/ divisional databases

Enterprise database

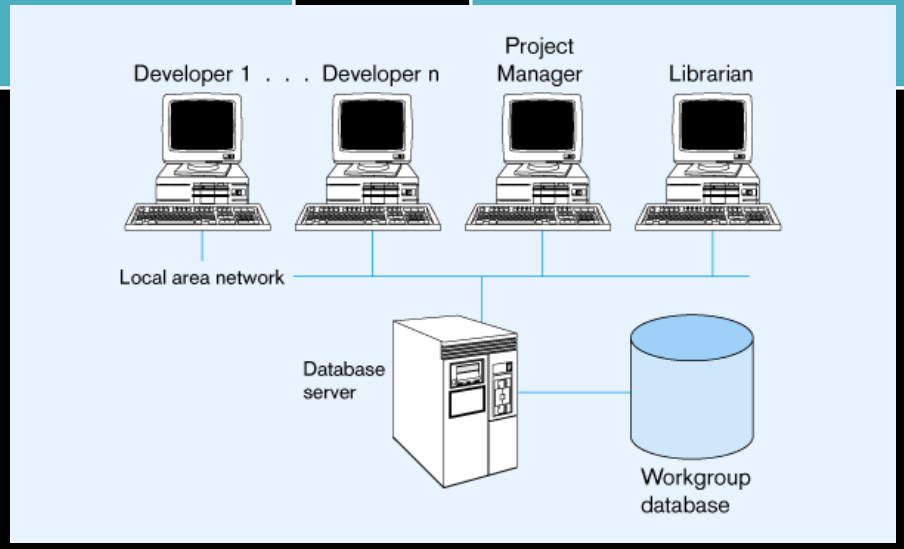
- Enterprise resource planning (ERP) systems
- Data warehousing implementations

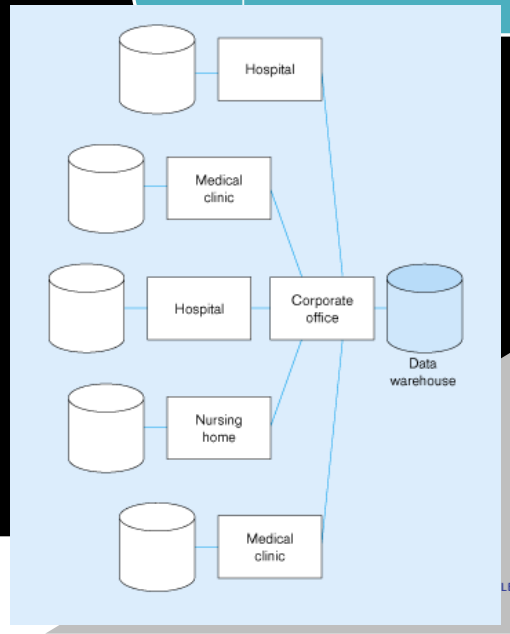
Customer

Customer Name: **Multi Media, Inc.**
Address: **1000 River Road**
City: **San Antonio**
State: **TX**
Zip: **76235**
Phone: **(219) 864-2000**
Next Contact Date: **10/17/2000** / Time: **10:30 AM**

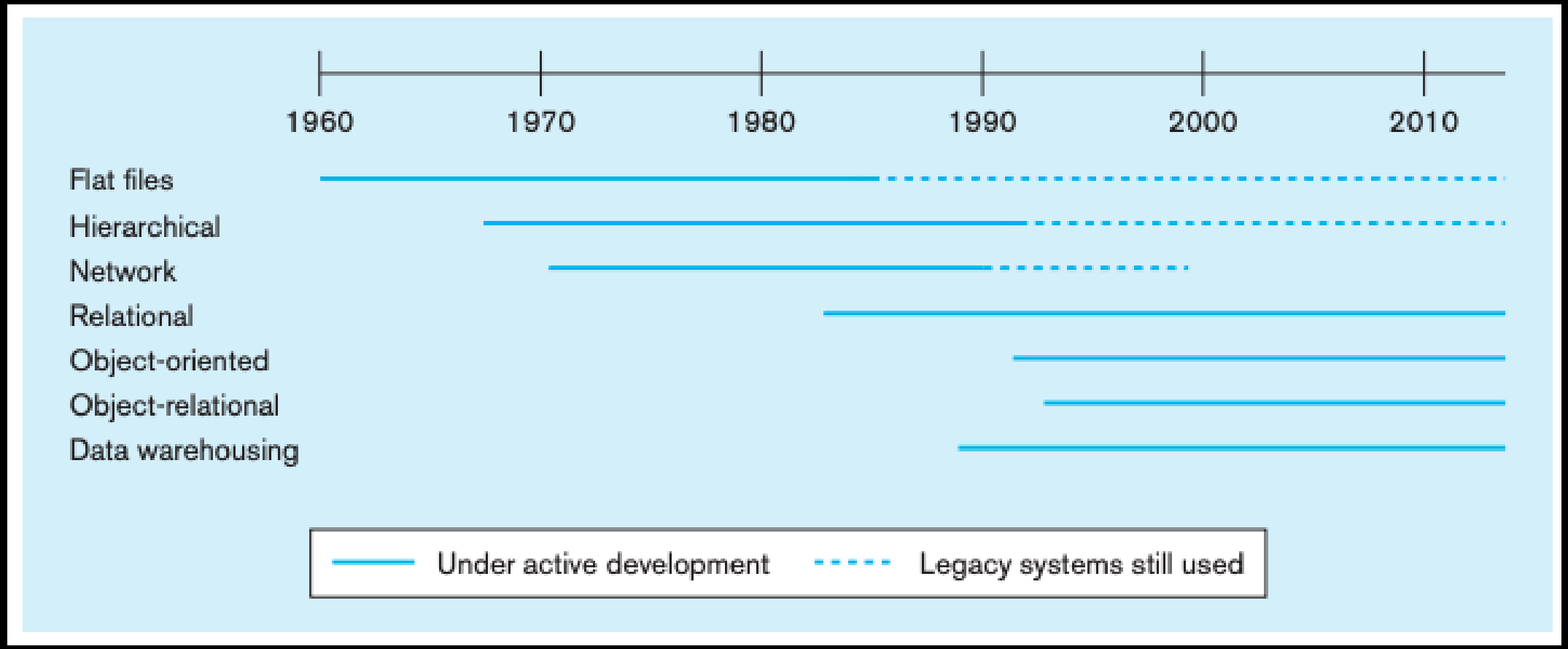
Contact History for Customer

Date	Time	Contact	Comments
08/04/2000	10:00 AM	Roberts	Review proposal
08/19/2000	08:00 AM	Roberts	Revise schedule
09/10/2000	09:00 AM	Pearson	Sign contract
09/21/2000	02:00 PM	Roberts	Follow up





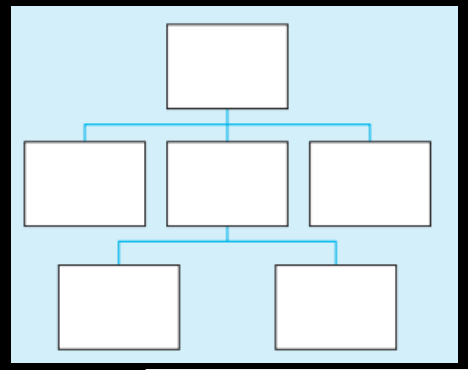
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Evolution of Database Technologies

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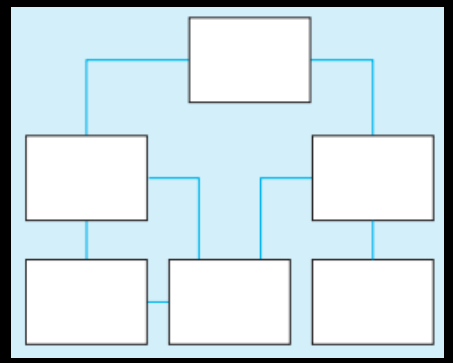
Various Data Models



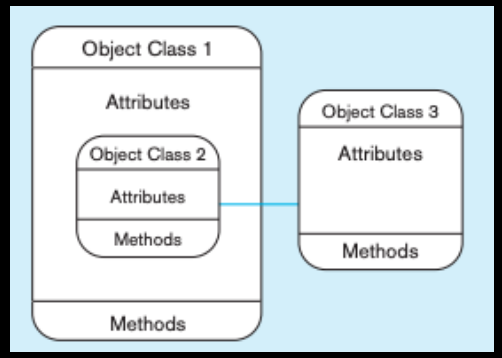
Hierarchical database model



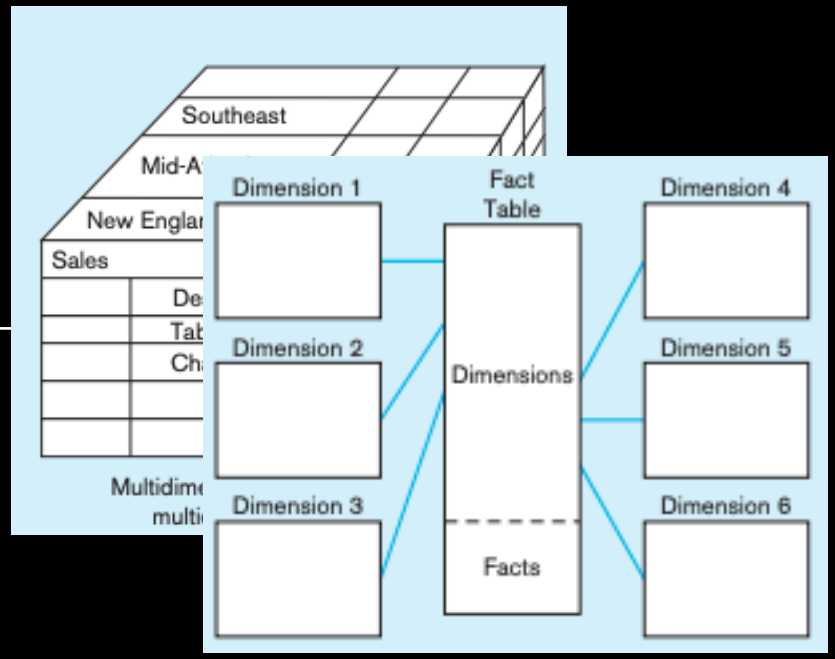
Relational database model



Network database model



Object-oriented database model



Multidimensional database model

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Levels of Abstraction

Physical level: describes how a record is stored.

Logical level: describes data stored in database, and the relationships among the data.

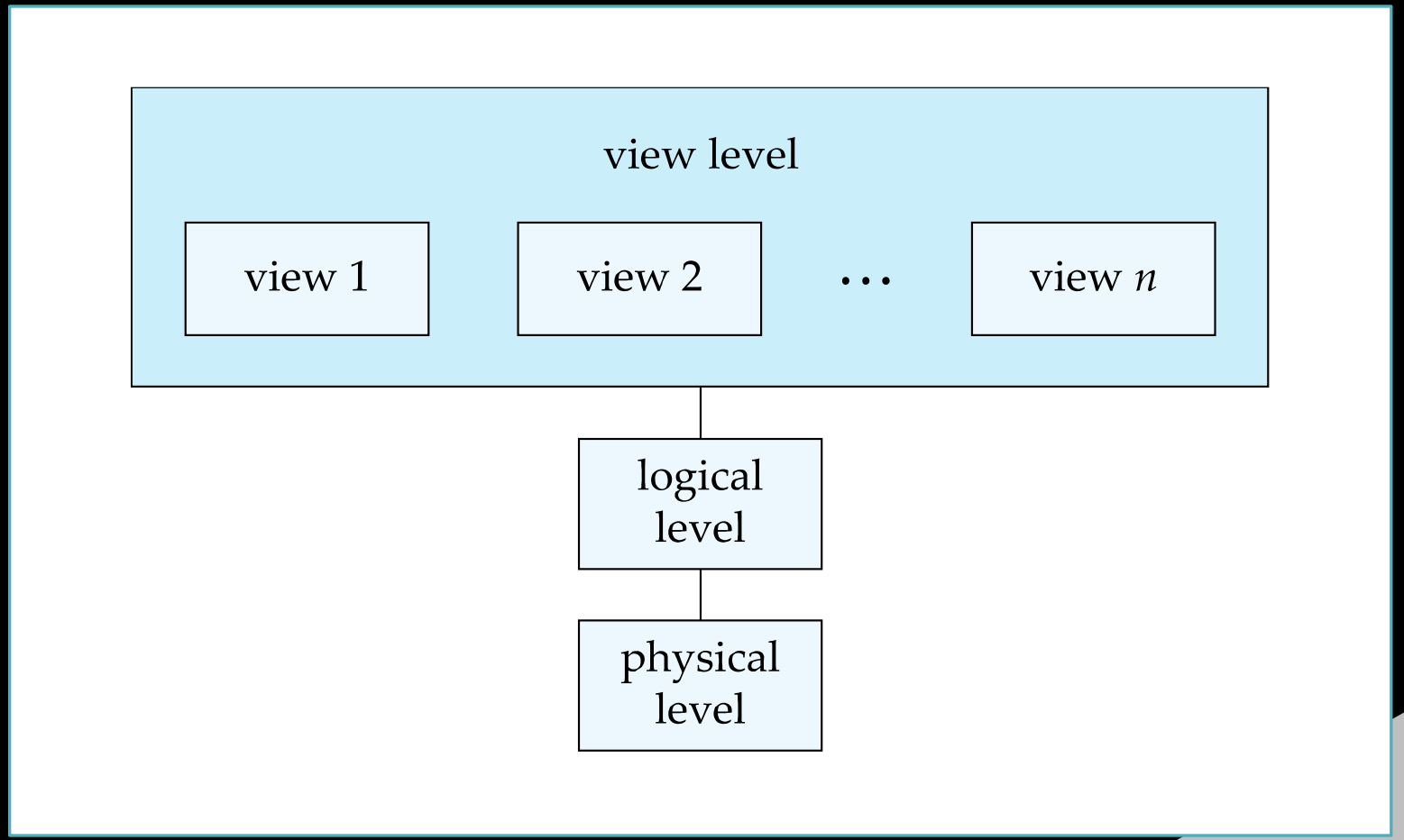
```
type instructor = record
        ID : string;
        name : string;
        dept_name : string;
        salary : integer;
    end;
```

View level: application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

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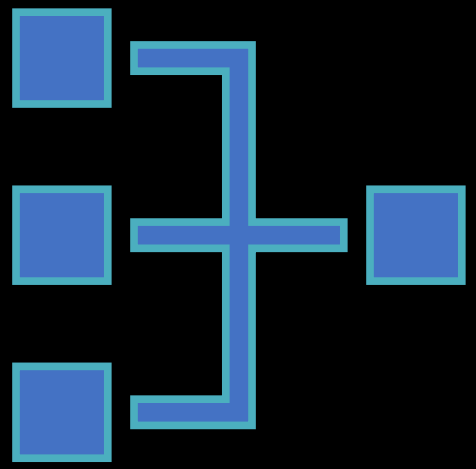
View of Data

An architecture for a database system (ANSI/SPARC Architecture)



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Physical Data Independence



Physical Data Independence – the ability to modify the physical schema without changing the logical schema

- Applications depend on the logical schema
- In general, the interfaces between the various levels and components should be well defined so that changes in some parts do not seriously influence others.

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Instances and Schemas



Schema : structure of the database

Logical Schema – the overall logical structure of the database

Physical Schema – the overall physical structure of the database



Instance : the actual content of the database at a particular point in time

Similar to types and variables in programming languages

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Data Definition Language (DDL)

Specification notation for defining the database schema

Example:
```
create table instructor (
    ID          char(5),
    name        varchar(20),
    dept_name   varchar(20),
    salary      numeric(8,2))
```

DDL compiler generates a set of table templates stored in a ***data dictionary*** - contains metadata (i.e., data about data)

- Database schema
- Integrity constraints
  - Primary key (ID uniquely identifies instructors)
- Authorization
  - Who can access what

Data Manipulation Language (DML)

Language for accessing and updating the data organized by the appropriate data model

There are basically two types of data-manipulation language

- **Procedural DML** -- require a user to specify what data are needed and how to get those data.
- **Declarative DML** -- require a user to specify what data are needed without specifying how to get those data.

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SQL Query Language

SQL query language is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table.

Example to find all instructors in Comp. Sci. dept

```
select name
from instructor
where dept_name = 'Comp. Sci.'
```

Application programs generally access databases through one of

- Language extensions to allow embedded SQL
- Application program interface (e.g., ODBC/JDBC) which allow SQL queries to be sent to a database



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Database Design



Logical Design – Deciding on the database schema. Database design requires that we find a "good" collection of relation schemas.

Business decision – What attributes should we record?

Computer Science decision – What relation schemas should we have?



Physical Design – Deciding on the physical layout of the database

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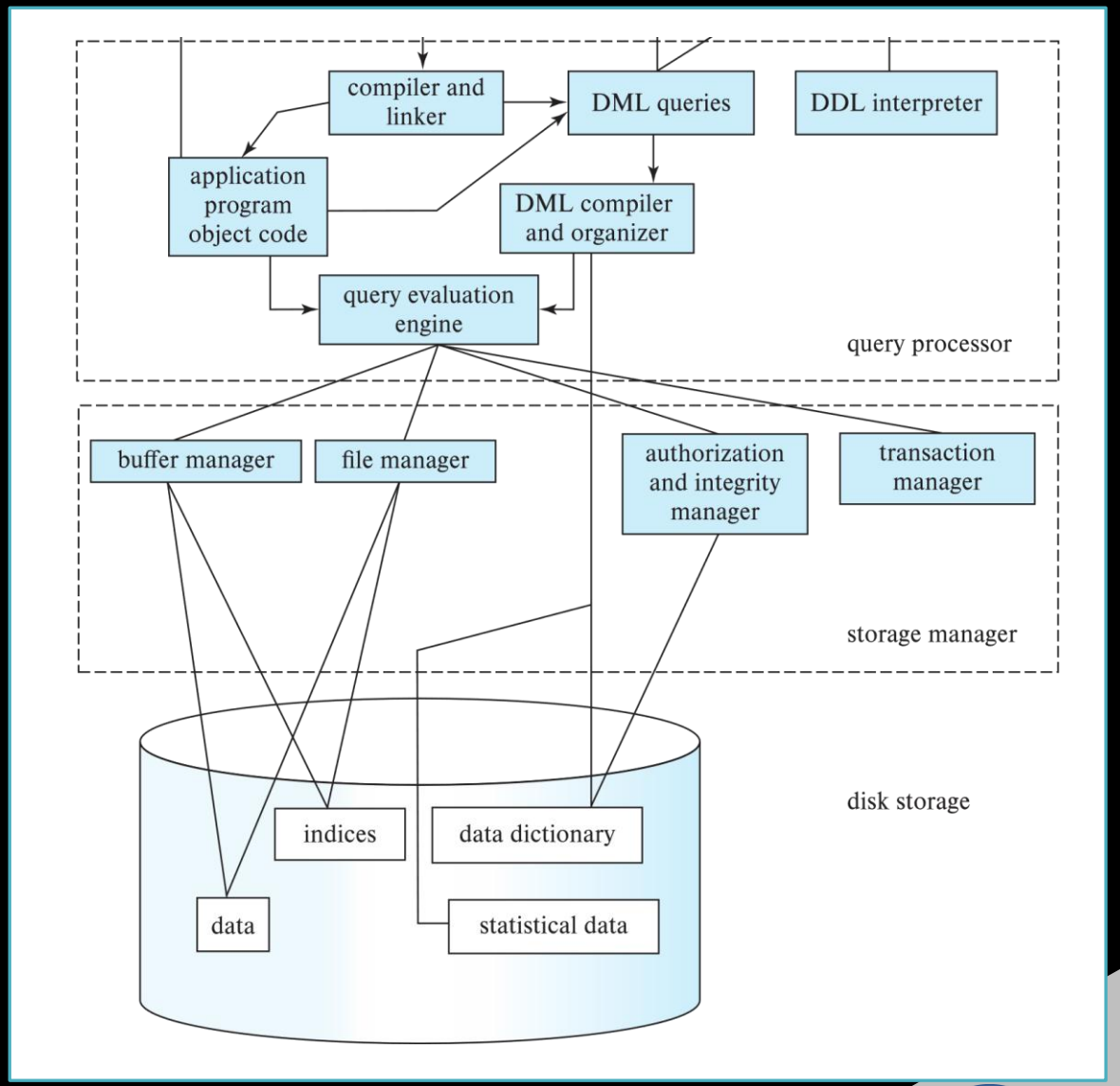
Database Engine

- A database system is partitioned into modules that deal with each of the responsibilities of the overall system.
- The **functional components** of a database system can be divided into:

The storage manager	The query processor component	The transaction management component

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Database Architecture

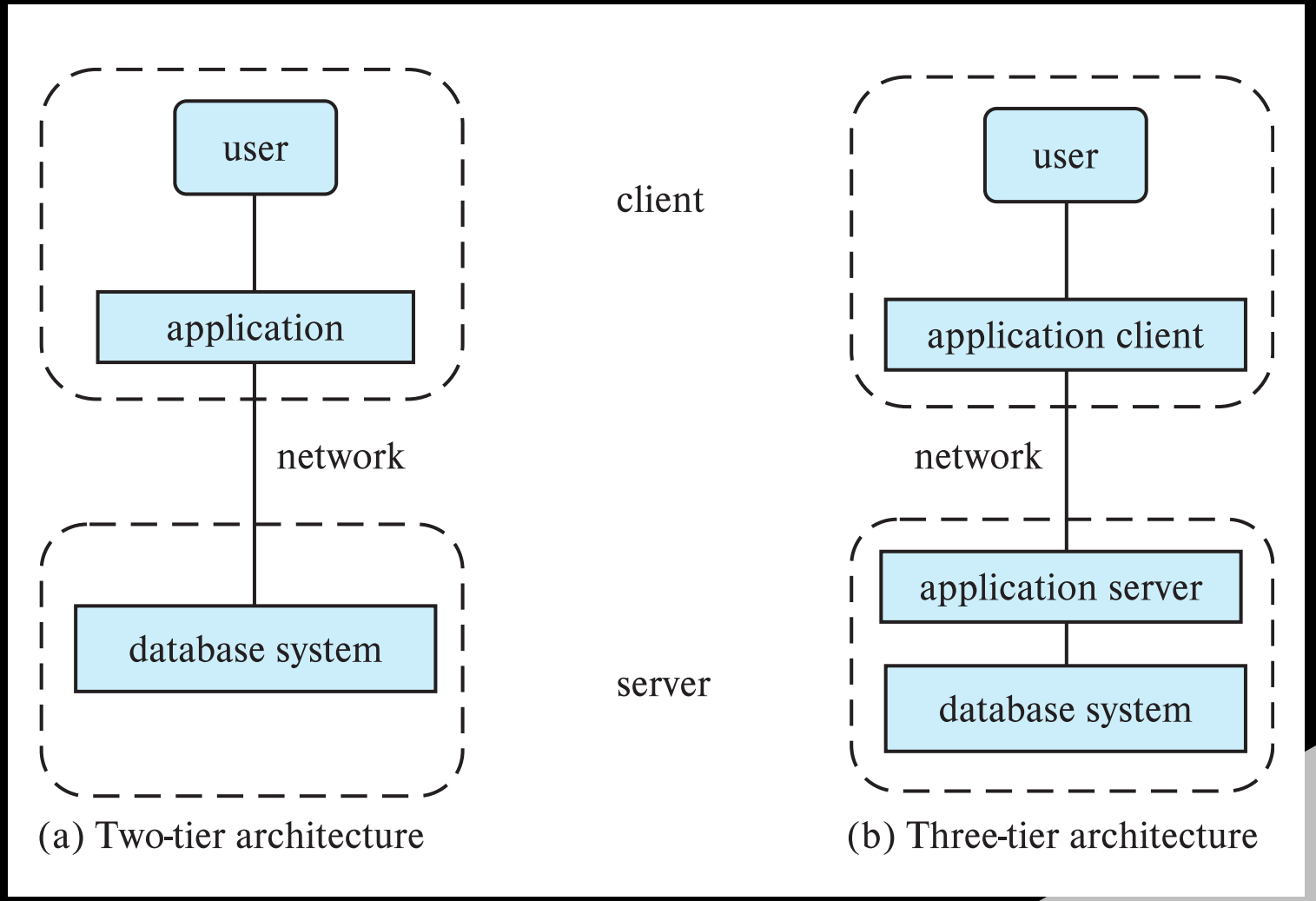


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Database Applications

Two-tier architecture -- the application resides at the client machine, where it invokes database system functionality at the server machine

Three-tier architecture -- the client machine acts as a front end and does not contain any direct database calls.

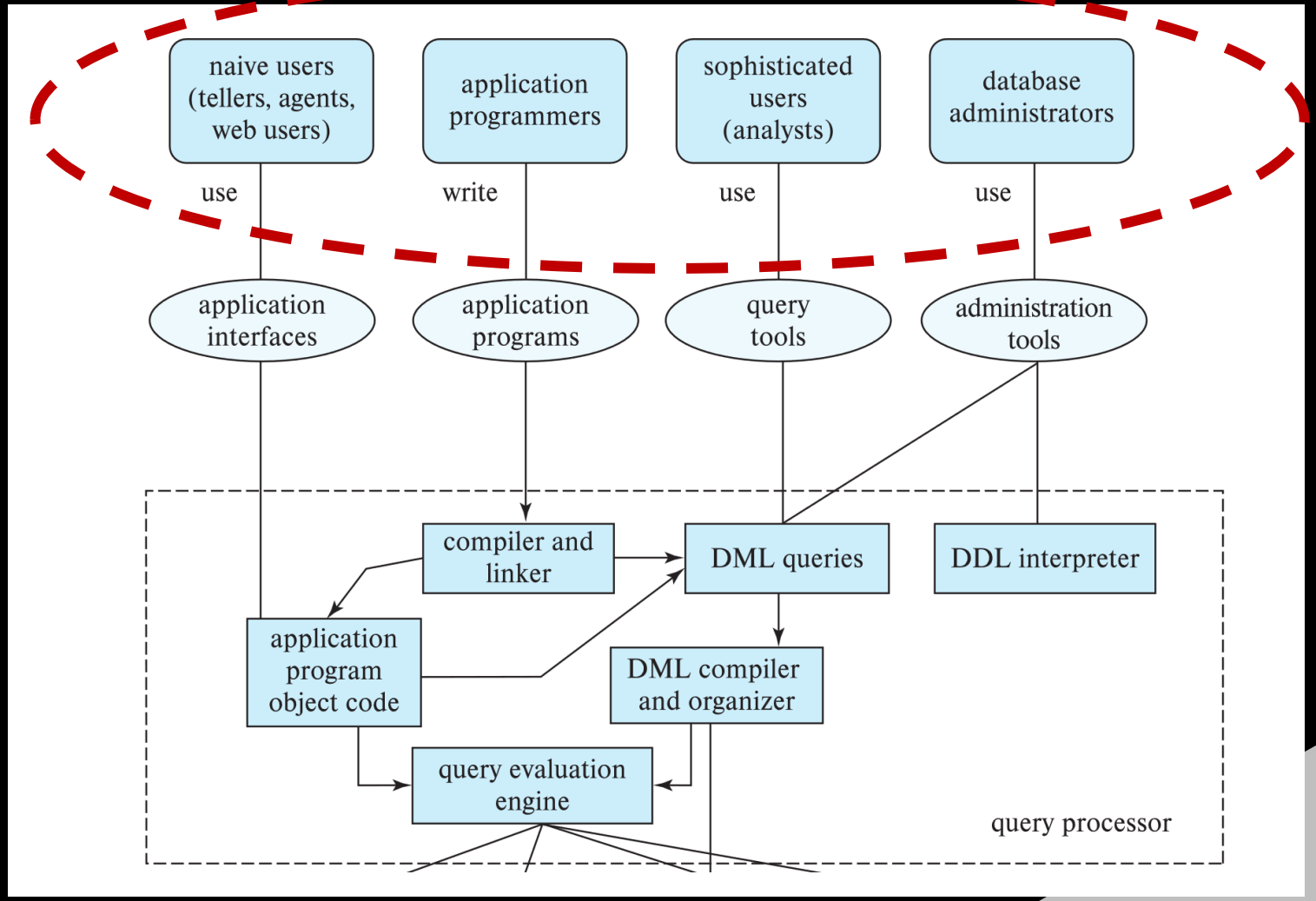


(a) Two-tier architecture

(b) Three-tier architecture

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Database Users



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